§ 6.1 Inner product, length, and orthogonality

Goal Put an extra structure on a vector space so that we can talk about distance

The inner product

$\vec{u}, \vec{v} \in \mathbb{R}^n$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \qquad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Def The inner product (or dot product) is

$$\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v} = [u_1 \; u_2 \; \cdots \; u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$= u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Exmp $\vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\vec{u} \cdot \vec{v} = 1 \times 0 + 0 \times 1 = 0$$

Exmp $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$$\vec{u} \cdot \vec{v} = 1 \times 3 + 1 \times 2 = 5.$$

Thm 1 $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$, $c \in \mathbb{R}$ scalar. Then

a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$ — Commutativity

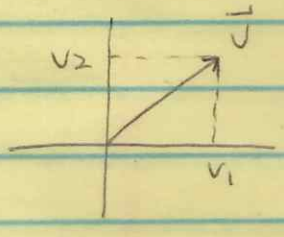
b) $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$ } linearity

c) $c(\vec{u} \cdot \vec{v}) = (c\vec{u}) \cdot \vec{v} = \vec{u} \cdot (c\vec{v})$ } linearity

d) $\vec{u} \cdot \vec{u} \geq 0$ and $\vec{u} \cdot \vec{u} = 0$ iff $\vec{u} = 0$ — positivity

The length of a vector

$$\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$



Def The length of $\vec{v}$ is

$$\|\vec{v}\| = \sqrt{\vec{v}\cdot\vec{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Notice that

- $\|\vec{v}\|^2 = \vec{v}\cdot\vec{v}$
- c scalar $\|c\vec{v}\| = |c|\,\|\vec{v}\|$ (absolute value)

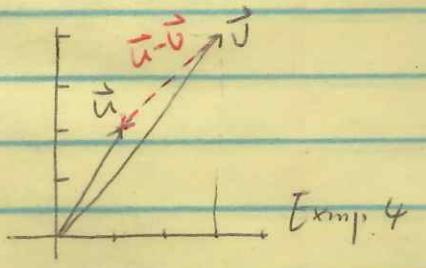
Def A vector of length 1 is a unit vector.

Suppose $\vec{v} \neq \vec{0}$, $\dfrac{\vec{v}}{\|\vec{v}\|}$ is a unit vector in the same direction as $\vec{v}$.

Exmp 2 $\vec{v} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix}$ Find a unit vector in the same direction as $\vec{v}$.

$$\|\vec{v}\| = \sqrt{\vec{v}\cdot\vec{v}} = \sqrt{1^2 + (-2)^2 + 2^2 + 0^2} = 3$$

$$\frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{3}\begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$



How we can talk about distance

Distance in $\mathbb{R}^n$

Def $\vec{u}, \vec{v} \in \mathbb{R}^n$ The distance between $\vec{u}$ & $\vec{v}$, written $\text{dist}(\vec{u},\vec{v})$

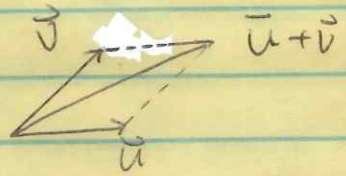
$$\text{dist}(\vec{u},\vec{v}) = \|\vec{u} - \vec{v}\|$$

Exmp 4 $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$$\text{dist}(\vec{u},\vec{v}) = \|\vec{u} - \vec{v}\| = \left\|\begin{bmatrix} -2 \\ -2 \end{bmatrix}\right\| = \sqrt{(-2)^2 + (-2)^2} = 2\sqrt{2}$$

Orthogonality

$\vec{u}, \vec{v}$ vectors in $\mathbb{R}^n$



$$\|\vec{u}+\vec{v}\|^2 = (\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v})$$
$$= \vec{u}\cdot(\vec{u}+\vec{v}) + \vec{v}\cdot(\vec{u}+\vec{v})$$
$$= \vec{u}\cdot\vec{u} + \vec{u}\cdot\vec{v} + \vec{v}\cdot\vec{u} + \vec{v}\cdot\vec{v}$$
$$= \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u}\cdot\vec{v}$$

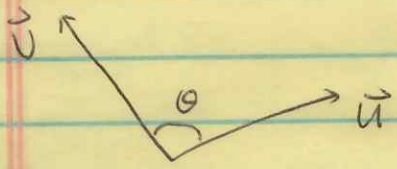
Thm 2 (Pythagorean Theorem) $\vec{u}$ and $\vec{v}$ are orthogonal iff

$$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$$

Thus we have the following definition

Def $\vec{u}$ and $\vec{v}$ are **orthogonal** if $\vec{u}\cdot\vec{v} = 0$.

Angles



The angle θ between vectors $\vec{u}$ & $\vec{v}$ is given by the formula

$$\vec{u}\cdot\vec{v} = \|\vec{u}\|\|\vec{v}\|\cos\theta.$$